Types of Graphs

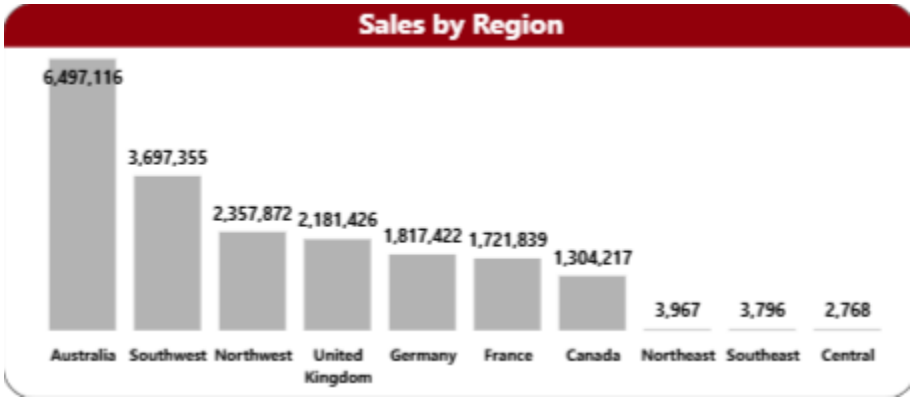
•Bar Graph

•Definition : Bar Graph is a type of graph that uses bars to represent data values. The length or height of each bar corresponds to the quantity it represents.

Bars can be displayed vertically or horizontally, and each bar represents a different category.

•When to use a Bar Graph:

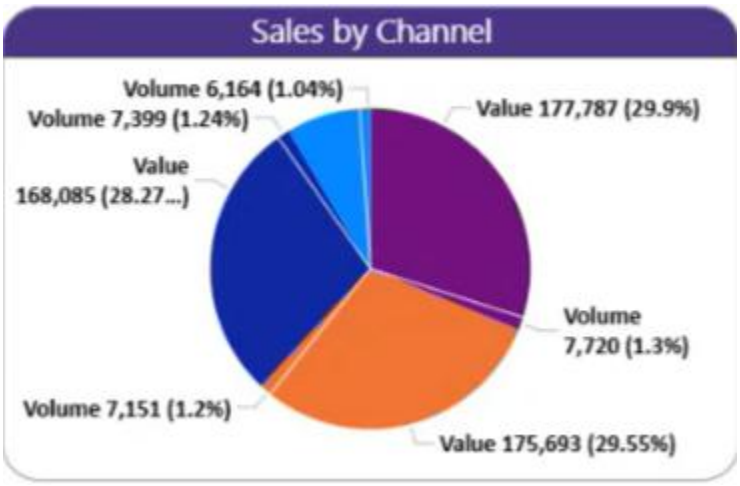
1- To compare quantities between different categories.
2- To show differences clearly and visually.
3- To display categorical data (such as subjects, products, or groups).
4- To make comparisons easy and straightforward.



•**Pie Graph**

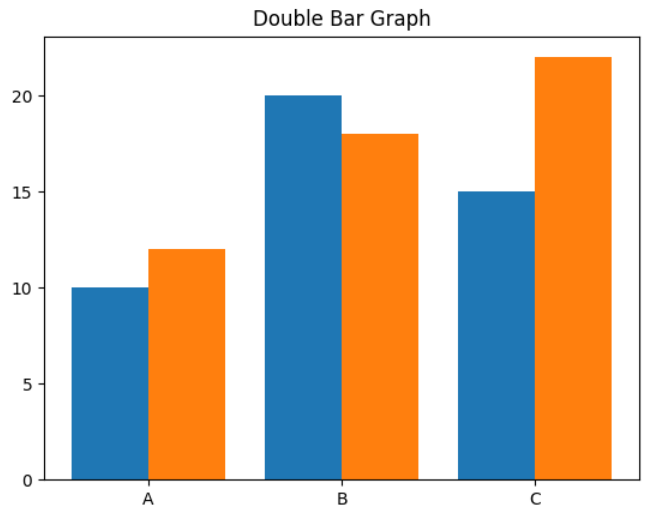
•Definition: is a circular chart that represents data as slices of a whole. Each slice shows a proportion or percentage of the total.

•When to use: Useful for showing percentage or proportional data.

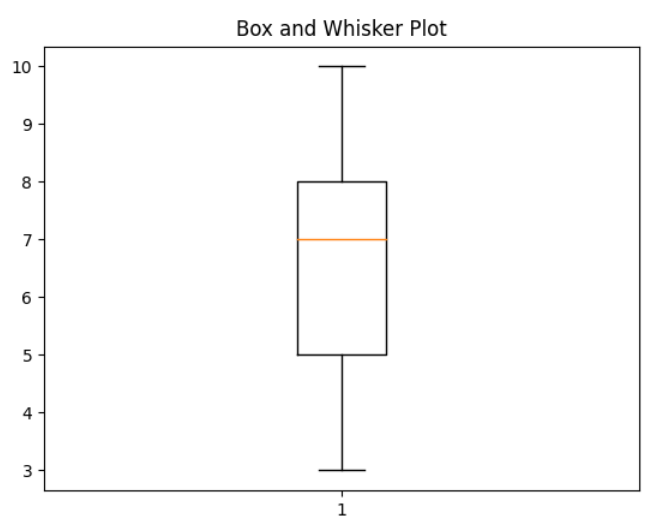


•**Double Bar Graph**

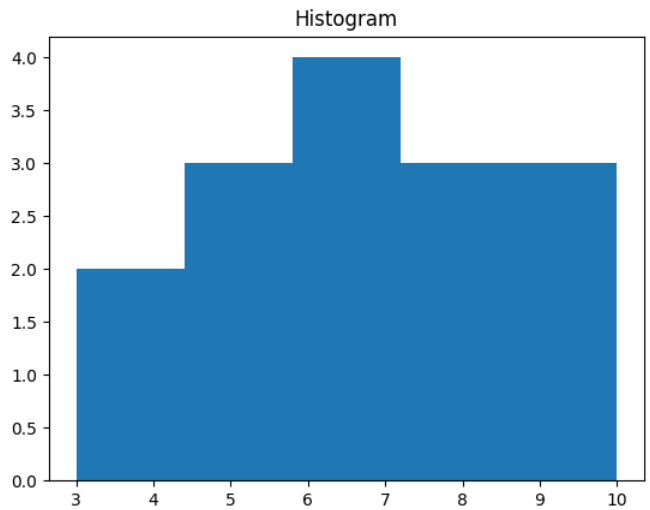
- Definition: A double bar graph compares two sets of data side by side.
- When to use: Used to compare two related groups across categories.



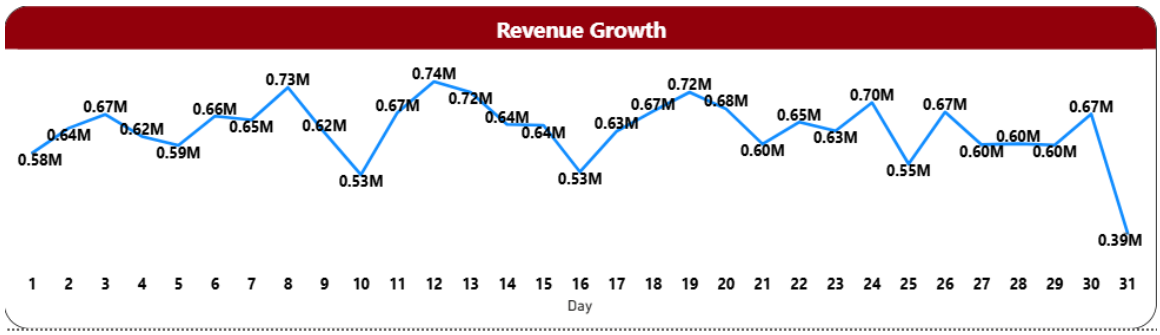
- **Box and Whisker Plot**
  - Definition: A box plot shows the distribution of data based on five-number summary.
  - When to use: Useful for identifying spread, median, and outliers.



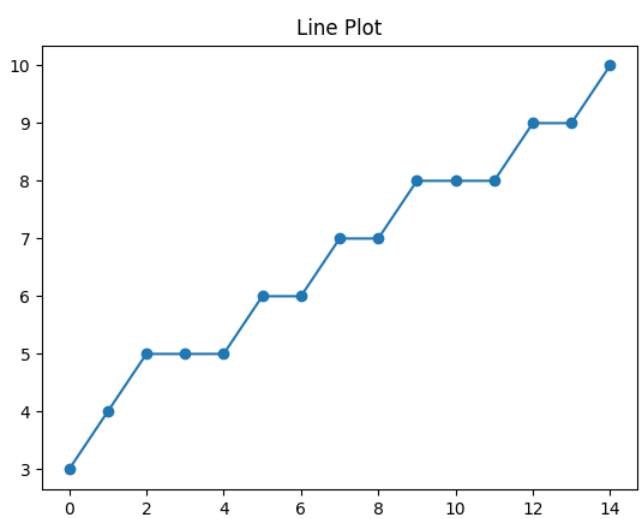
- **Histogram**
  - Definition: A histogram shows frequency distribution of continuous data.
  - When to use: Used to understand data distribution shape.



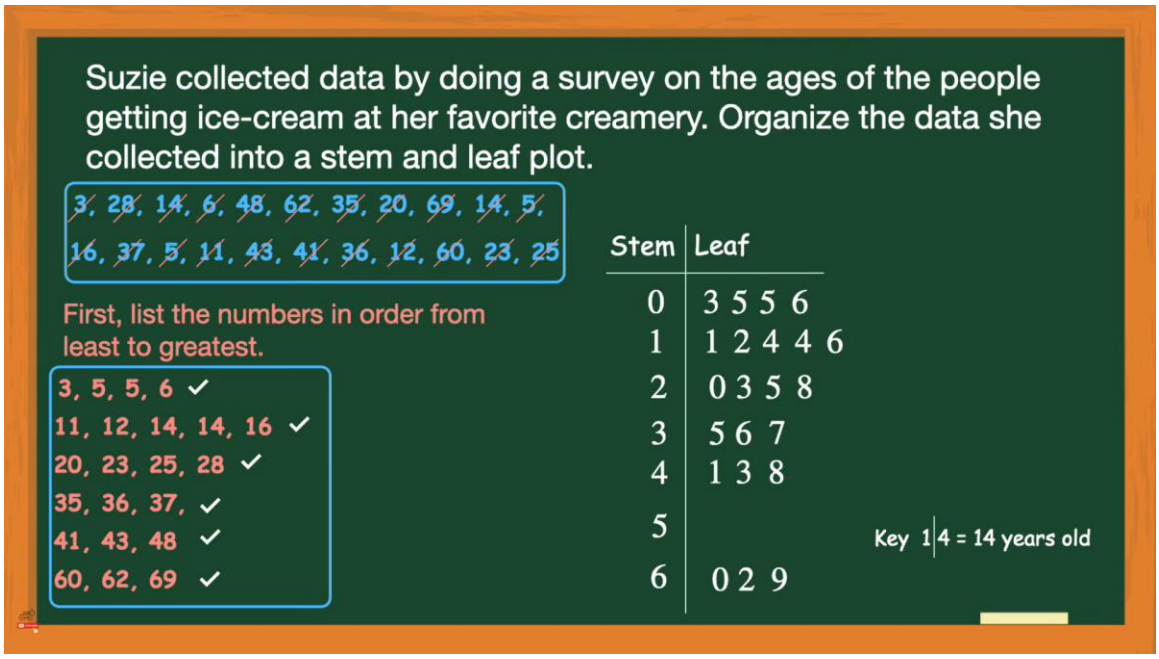
- **Line Graph**
  - Definition : is a type of graph that uses points connected by straight lines to show changes in data over time or across ordered categories
  - When to use: Useful for showing trends over time.



- **Line Plot**
  - Definition: A line plot displays data along a number line. Also line plot shows the frequency of data by stacking marks above the same value.
- When to use a Line Plot:
  - To display small data sets clearly.
  - To show the frequency of individual values.



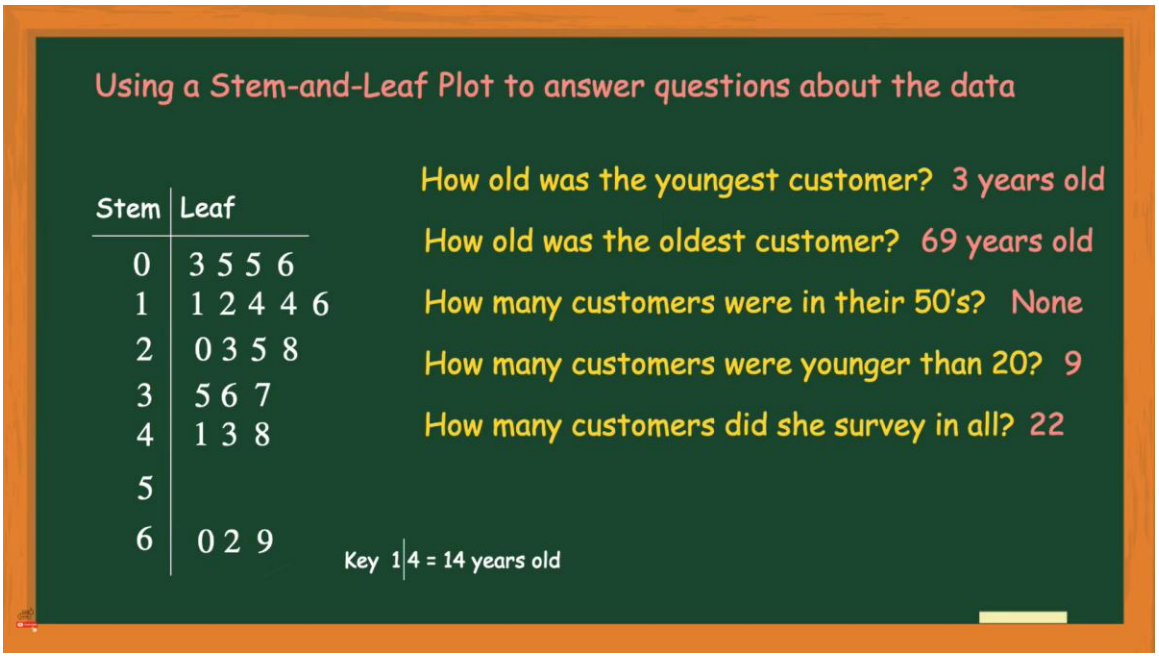
- **Stem and Leaf Plot**
  - Definition: A stem-and-leaf plot organizes data to show distribution.
  - When to use: Useful for displaying actual data values clearly.



As shown first we sort the data set

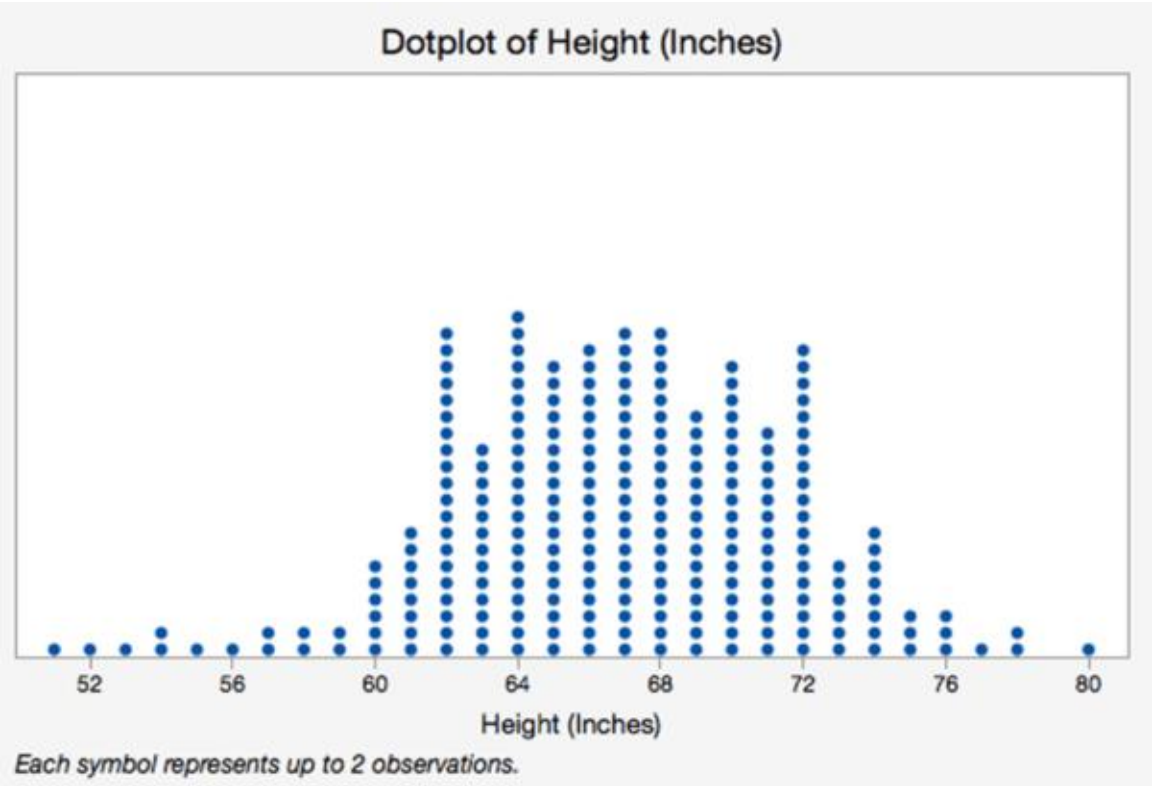
Then, we separate each number's digits so first part is stem(tens) and the last digit is leaf

Then we make stem/leaf table as shown and we pick a random number and use it as a key for clarity .



- This way makes it easier to understand the data and to be able to get answers for our questions

- **Dot Plot**
- Definition: A dot plot uses dots to represent data values.
- When to use: Useful for showing distribution of small data sets.



- **Symmetry and Skewness**
- Definition: Symmetry and skewness describe data distribution shape.
- Symmetry : means that the distribution of data is balanced around the center (usually the mean or median).
- Skewness : measures the degree and direction of asymmetry in a data distribution.
- When to use: Useful to understand whether data is balanced or skewed.

• Types of skewness:

Positive Skewness (Right-Skewed Distribution): The tail of the distribution extends to the right. Most data values are concentrated on the left, with a few unusually large values. In this case, the mean is greater than the median.

Negative Skewness (Left-Skewed Distribution):The tail of the distribution extends to the left. Most data values are concentrated on the right, with a few unusually small values. Here, the mean is less than the median

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Heatmap

- Definition: represents values using colors. The intensity or shade of the color indicates the magnitude of the data.
- A heatmap uses a color scale (for example, light to dark or blue to red) to show low and high values clearly.
- When to use a Heatmap:
- To visualize patterns and trends in large datasets.

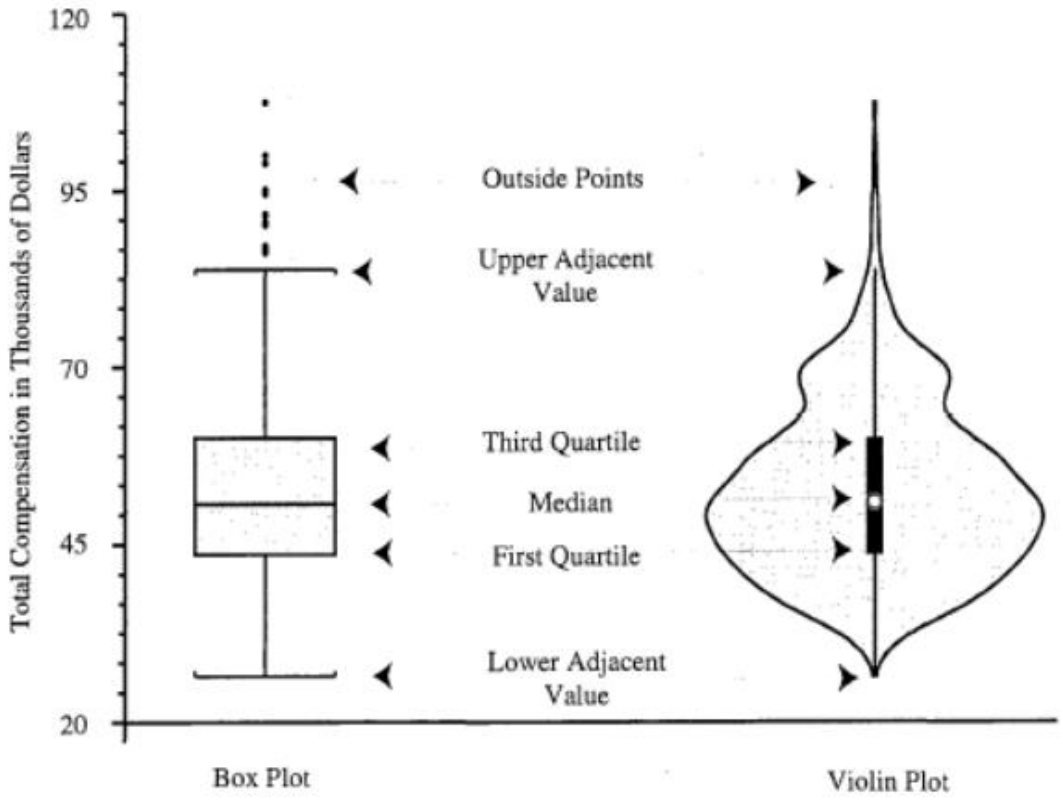
Product Performance

Category	NewFamily	Item Name	YTD	Yago	%Yago	YTD Var	MTD	Mago	%Mago	Mago Var	Sales I
Chocolate	Corona M	Chocolate Dark Med.	1,183,846	3,394,318	-65%	-2,210,472	303,711	123,931	145%	179,780	-18,12
Chocolate	Corona M	Chocolate Milk Med.	960,736	3,377,164	-72%	-2,416,429	180,826	-21	-864981%	180,847	-6,44
Chocolate	Corona M	Chocolate Crispy Med.	489,956	1,621,495	-70%	-1,131,539	179,905	0	-100%	179,905	0
Chocolate	Corona M	Chocolate Extra Milk Med.	0	0		0	0	0	-100%	0	0
Chocolate	Corona M	Total	2,634,538	8,392,977	-69%	-5,758,439	664,442	123,911	436%	540,532	-24,56
Chocolate	New Range Dark L	Chocolate Dark Hazelnut ...	1,564,259	2,041,059	-23%	-476,801	199,504	29,659	573%	169,846	-4,62
Chocolate	New Range Dark L	Chocolate Dark Stevia 55%	1,816,893	1,505,873	21%	311,020	324,952	175,867	85%	149,085	-41
Chocolate	New Range Dark L	Chocolate Dark Almond ...	921,127	1,310,219	-30%	-389,092	286,966	127,471	125%	159,495	-286
Chocolate	New Range Dark L	Chocolate Dark 72%	2,326,441	1,053,830	121%	1,272,611	583,190	82,472	607%	500,718	-82
Chocolate	New Range Dark L	Total	6,628,720	5,910,982	12%	717,738	1,394,613	415,470	236%	979,143	-5,03
Chocolate	Corona L	Chocolate Extra Milk Large	806,889	1,936,128	-58%	-1,129,238	146,615	-1,527	-9700%	148,142	-986
Chocolate	Corona L	Chocolate Crispy Large	1,004,539	1,874,542	-46%	-870,003	133,760	121,988	10%	11,772	-3,85
Chocolate	Corona L	Chocolate Milk Large	-17,150	977,032	-102%	-994,181	0	0	-100%	0	0
Chocolate	Corona L	Chocolate Hazelnut Large	0	209,262	-100%	-209,262	0	0	-100%	0	0
Chocolate	Corona L	Total	1,794,279	4,996,964	-64%	-3,202,685	280,375	120,461	133%	159,914	-4,83
Chocolate	Corona Couverture	Couverture Dark Med. 1...	2,147,877	2,102,137	2%	45,740	759,147	-360	-210960%	759,507	-2,08
Chocolate	Corona Couverture	Total	2,147,877	2,102,137	2%	45,740	759,147	-360	-210960%	759,507	-2,08
Chocolate	New Range Milk L	Chocolate Milk Salted Al...	904,412	1,451,782	-38%	-547,370	-10,229	103,090	-110%	-113,319	-1,84
Chocolate	New Range Milk L	Chocolate Milk Stevia Lar...	-3,396	422,184	-101%	-425,580	0	170,836	-100%	-170,836	0
Chocolate	New Range Milk L	Total	901,016	1,873,966	-52%	-972,950	-10,229	273,926	-104%	-284,155	-1,84
Chocolate	Fun Fruit	Fun Fruit Orange	-5,164	403,100	-101%	-408,264	0	0	-100%	0	0
Chocolate	Fun Fruit	Fun Fruit Strawberry	-293	29,454	-101%	-29,747	0	0	-100%	0	0
Chocolate	Fun Fruit	Total	-5,457	432,555	-101%	-438,011	0	0	-100%	0	0
Chocolate	Groovy	Groovy Dark Chocolate L...	501,813	0		501,813	501,813	0	-100%	501,813	0

- **Violin Plot**
- Definition: is a data visualization that combines a box plot and a density plot. It shows the distribution, probability density, median, and interquartile range of a dataset. The width of the violin at different values indicates the density of data points wider sections mean more data points at that value.

- When to use:

- 1- To visualize the distribution and spread of data.
- 2- To compare distributions between multiple groups.
- 3- To identify the median, quartiles, and density simultaneously.
- 4- To detect multimodal distributions (data with multiple peaks)


120
95
70
45
20
Total Compensation in Thousands of Dollars
Outside Points
Upper Adjacent Value
Third Quartile
Median
First Quartile
Lower Adjacent Value
Box Plot
Violin Plot